W1 – Equations of Lines in Two-Space Unit 6

MCV4U

Jensen

1) Write a vector equation for a line given each direction vector $\vec{m}$ and point P_0.

a) $\vec{m} = [2, -7]$, $P_0(9,4)$ **b)** $\vec{m} = [-1, 5]$, $P_0(3,7)$

2) Write the vector equation of the line that passes through each pair of points.

a) $A(2,3)$, $B(7,1)$ **b)** $A(5, -1)$, $B(-2,7)$

3)Write the parametric equations for each vector equation.

a) $[x, y] = [7,3] + t[2,5]$ **b)** $[x, y] = [-1,4] + t[8, -9]$

4) Determine if each point P is on the line $[x, y] = [3,1] + t[-2,5]$

a) P**(**12, 36) **b)** P(–7, 26)

c) P(1, 6) **d)** P(–3, 15)

5) Write a vector equation for each line, given the parametric equations.

a) $\ell: \begin{cases} x = 2 - 4t \\ y = 1 - 5t \end{cases}$

b) $\ell: \begin{cases} x = 8 - 9t \\ y = 3 \end{cases}$

6) Given each set of parametric equations, write the scalar equation.

a) $\ell: \begin{cases} x = 3 + 4t \\ y = -1 - 5t \end{cases}$

b) $\ell: \begin{cases} x = 2 + 8t \\ y = -5 + 7t \end{cases}$

7) Write the scalar equation of each line given the normal vector $\vec{n}$ and point P_0.

a) $\vec{n} = [4,1]$, $P_0(3, -5)$

b) $\vec{n} = [1, -3]$, $P_0(4,3)$

8) Determine the vector equation of each line.

a) parallel to the *y*-axis and through $P_0(2, -7)$

b) parallel to $[x, y] = [3, -1] + t[3, 4]$ and through $P_0(-2, 4)$

c) perpendicular to $[x, y] = [2, -1] + t[5, 3]$ with *x*-intercept 3

9) Given each scalar equation, write a vector equation and the parametric equations.

a) $x + 2y = 6$

b) $5x - 2y = 13$

10) In each case, determine if ℓ_1 and ℓ_2 are parallel, perpendicular, or neither. Explain.

a) $\ell_1: 4x - 6y = 9$

$\ell_2: [x, y] = [6,3] + t[3,2]$

b) $\ell_1: x + 9y = 2$

$\ell_2: \begin{cases} x = t \\ y = 15 + 9t \end{cases}$

Answers:

1)a) [*x*, *y*] = [9, 4] + *t*[2, –7] **b)** [*x*, *y*] = [3, 7] + *t*[–1, 5]

2)a) $[x, y] = [2,3] + t[5, -2]$ **b)** $[x, y] = [5, -1] + t[-7,8]$

3)a) $\ell: \begin{cases} x = 7 + 2t \\ y = 3 + 5t \end{cases}$ **b)** $\ell: \begin{cases} x = -1 + 8t \\ y = 4 - 9t \end{cases}$

4)a) No **b)** Yes **c)** Yes **d)** No

5)a) [*x*, *y*] = [2, 1] + *t*[–4, –5] **b)** [*x*, *y*] = [8, 3] + *t*[–9, 0]

6)a) $5x + 4y - 11 = 0$ **b)** $-7x + 8y + 54 = 0$

7)a) $4x + y - 7 = 0$ **b)** $x - 3y + 5 = 0$

8)a) [*x*, *y*] = [2, –7] + *t*[0, 1] **b)** [*x*, *y*] = [–2, 4] + *t*[3, 4] **c)** [*x*, *y*] = [3, 0] + *t*[–3, 5]

9)a) $[x, y] = [2,2] + t[-2,1]; x = 2 - 2t, y = 2 + t$ **b)** $[x, y] = [3,1] + t[2,5]; x = 3 + 2t, y = 1 + 5t$

10)a) parallel **b)** perpendicular